CHAPTER CONTENT FOR BSCS

Alon: Audio Speech Evaluator and Recommendation System for Effective Communication

A Research Project

Presented to the

Faculty of the College of Computer Studies and Systems

University of the East, Manila

In Partial Fulfillment

Of the Requirements for the Degree in

Bachelor of Science in Computer Science

By

Von Andrei C. Toledo

Chris-Ivan C. Maghirang

Aica D. Ortega

Mark Elwyn L. Espana

Laureano Vincent L. Palanas

<< Date >>

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Chapter 1

THE PROBLEM AND ITS BACKGROUND

Introduction

Introduction gives the readers a panoramic view of the whole study. It usually discusses the research area of the study. It should be proportionate to the length of the whole paper (usually, one-page introduction is enough).

Background of the Study

This section discusses what the study is all about and should clearly present the gap of the previous studies. This allows the researchers to clearly state their contribution to the body of knowledge.

Statement of the Problem

This section identifies the research problems to be investigated. The manner of stating the problems is dependent on the type of research designs the researcher intended to use. The Statement of the Problem should in question form. The general and specific statements of the study should be presented.

For example:

The study aims to determine the factors that affect web-portal usability that can serve as inputs UE's faculty web-portal development. Specifically, the study intends to answer the following questions:

1. *What is the faculty-related factors in terms of:*
    a. *Age*
    b. *Highest Educational Attainment*
    c. *Level of Computer Applications and Internet Skills*
    d. *Commitment of Use Faculty Web Portal, and*
    e. *Presence of Internet Access at Home?*
2. *How evident are the following web-portal design-related factors in the existing faculty web-portal of UE as perceived by the respondents?*
    a. *Ease of Navigation*
    b. *Information Content*
    c. *Availability*
    d. *Speed, and*
    e. *Aesthetics*
3. *Do the Faculty-Related Factors and Web-Portal Design-Related Factors, singly or in combination, affect the faculty web-portal usability?*

Scope and Limitations of the Study

This portion discusses the extent of the study. This contains the necessary areas that the study would reach into. It discusses in detail the boundary of the study and the justification for such scope and limitation. It answers and presents the 5Ws and 1H questions (What? When? Where? Who? Why? and How?) of your study.

Significance of the Study

This portion notes the contribution of the proposed study. It expresses the benefit of the study to direct stakeholders.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the review of literature and existing studies relevant to the development of EquiVox: An Audio Speech Evaluator and Recommendation System for Effective Communication. In a logical sequence, topics are discussed on the degree of importance and relationship. It should be noted that all citations should be done in APA format.

You should come up with a logical outline of topics/areas related to the study topic. After conducting the necessary research, synthesize your content and present it logically. There should be smooth transition from one topic to another. The following are the required sources:-

Review of Related Literatures

Types of Source	Minimum Quantity
Books	10
Periodicals (Magazine, Article, Newspaper)	10

Review of Related Studies

Acoustic Features for Stress and Anxiety Detection

Speech contains various acoustic characteristics that can provide information about a speaker's emotional and physiological state. Among the commonly analyzed acoustic features are fundamental frequency, jitter, shimmer, intensity, and other low-level descriptors. These characteristics can change depending on a person's emotional condition, making speech analysis a potential non-invasive approach for identifying stress and anxiety.

Pallewela et al. (2024) investigated the use of advanced audio cues for speech emotion recognition through machine learning. Their study examined acoustic characteristics such as fundamental frequency, jitter, and shimmer and demonstrated their usefulness in identifying emotional intensity in speech. The findings suggest that measurable variations in vocal characteristics can provide meaningful information regarding a speaker's emotional state. This is relevant to the present study because jitter and shimmer can potentially serve as objective acoustic indicators when analyzing speech associated with anxiety or stress.

Vocal tremors and changes in vocal stability may also occur when an individual experiences heightened emotional pressure. During situations such as public

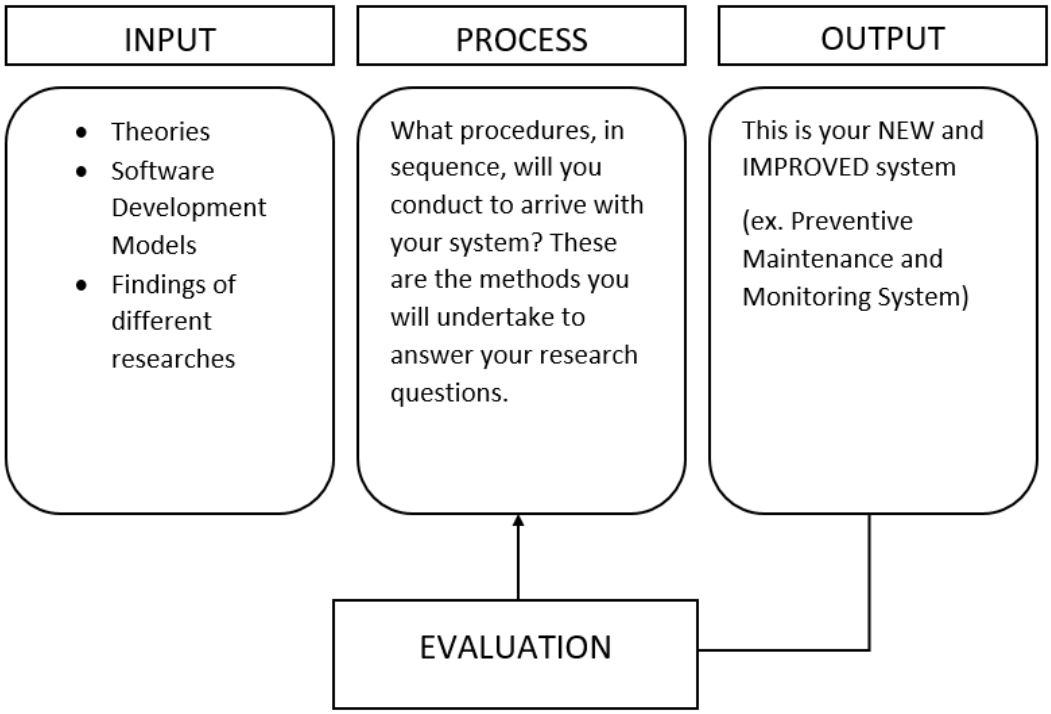
speaking, anxiety may affect voice production, resulting in noticeable changes in pitch, intensity, and vocal steadiness. These observations provide a basis for incorporating acoustic analysis into systems designed to identify possible anxiety-related speech patterns.

Furthermore, Murali et al. (2022) developed a real-time public speaking anxiety prediction model that utilized audio features to predict moments of anxiety during oral presentations. Their research is particularly relevant because it demonstrates the feasibility of analyzing speech over time rather than treating an entire presentation as a single observation. Such an approach supports the concept of identifying specific periods in a presentation when anxiety indicators become more prominent.

Types of Source	Minimum Quantity
Local Theses/Dissertations	5
Foreign Theses/Dissertations	5

Conceptual Framework

This section shows the framework that identifies various components of the study and depicts the relationship of the elements. The diagram below is a sample of a research framework for the study and explains the detail of each component. It must be noted that the Conceptual Framework of the study should conform to the Research Design of the study.



Hypothesis

This section states the null and alternative hypothesis that the study wants to investigate. For example:

H0: The Faculty-related and Web-portal design-related factors do not significantly affect faculty web-portal usability.

H1: The Faculty-related and Web-portal design-related factors significantly affect faculty web-portal usability.

Operational Definition of Terms

Define terminologies based on how they are used in the study. Terminologies are alphabetically arranged. The definition of each term is presented in a paragraph format, with the first line of the definition, indented. (The different criteria in the evaluation sheet should also be defined here.)

Chapter 3 – METHODOLOGY

This chapter should be consulted to the adviser because of the type of research the students will undergo. However, for Implementation-driven Research, students should include Software Design and Development Approach of the Software Development Methodology of the BSIT Software Project format.

Software Design and Development Approach

- Software process model *discusses the activities/procedures necessary to come up with the system. It shows and discusses the chosen software process model. Examples of such models are: waterfall model, prototype, RAD, Spiral model, etc. Steps in details should be discussed very well and a justification on the choice of process model should be included.*
- System Framework which may be a block diagram, *representing the major components of the system and their relationship with one another.*

*Note: In the System Framework, include the discussion of the scope of the software.

- Process Flow (HIPO and System Flowchart)

Research Design

Research design determines which statistical tool that the researcher has the use. There are numerous research designs. The researcher can choose in any of the following research designs: Descriptive, Experimental, Quasi-experimental and Correlational (Dancey and Reidy, 2002).

Determination of Sample Size

This section should present the method of getting a representative of a population.

Sampling Design and Technique

This section presents what type of sampling design and technique will be employed: Random, Stratified, Systematic, Multistage, Cluster, Purposive, Incidental, and Quota sampling (Calmorin and Calmorin, 1995).

The Subjects

This section describes who the foci of the study are.

Statistical Treatment of Data

This section presents the statistical tools that will be used in the study.

For CCS 4980 (THESA), Chapters 1 - 3 are initially written in either present or future verb tense. In the Final Paper, the tenses should be changed into past tense.

Chapter 4 – RESULTS AND DISCUSSIONS

Developed Prototype

Provide a thorough description of your prototype.

Features and Capabilities

- It enumerates and discusses the features of the prototype. What can it do? What are its advantages?
- It shows the screenshots of the various modules/interfaces (per user if applicable). A thorough explanation of the screenshot should be included.
- It discusses the result of the project in accordance with the objectives of the project.
- Screenshots of outputs (reports) with accompanying explanations, if applicable

Chapter 5 – SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

Summary of Findings

This section summarizes the findings of the paper.

Conclusions

This section discusses the interpretation of your research results in relationship with your statement of the problem.

- Were you able to achieve these objectives?
- What does the result of the evaluation procedure say about your study?

Recommendations

This section should answer the following questions:

- What modifications/improvements can you give to future researchers in reference to the weaknesses/limitations of the study?

Bibliography

This section should identify the different sources of information utilized in the accomplishment of the study. All sources mentioned in the document should be reflected here. Sources are categorized according to type (books, journal, theses, etc.) and should be written alphabetically according to the Author's Surname. Use the appropriate APA format for proper bibliographical entries.

Appendices

This section allows the group to include the documents related in their topic. This may include sample documents, sample screen outputs, interview, survey instruments, etc.

REFERENCES